

FiberMargin: Two-Sided Path Enclosures for Multiclass Spatial Label Repair

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Abstract

Multiclass spatial label fields can contain isolated errors, unstable borders, and mislabeled patches. We study training-free repair from coordinates and one imperfect categorical field, without images, logits, anchors, or reference labels. FiberMargin is a deterministic two-sided enclosure operator. Rotated Hilbert paths support forward and reverse exponential recurrences; adding their square-root amplitudes rewards rival-class evidence that encloses a query from both directions. A label changes only when the resulting atlas contrast clears a barrier determined by path disagreement, class prevalence, and local isolation. We derive an exact repair–damage decomposition, characterize the enclosure field, and establish finite-input stability.

Against eight equal-input controls, FiberMargin obtained the highest Accuracy and adjusted Rand index on 708 complete simulations, 108 unseen geometries, procedural multiclass masks, DLPFC, and MERFISH, and the highest Accuracy on variable-density 3D fields. It narrowly trailed multiscale mode on CRC. On saved CamVid and S3DIS predictions it improved the upstream labels but did not lead the comparison, delimiting its transfer range. The C++ implementation processed 500,000 observations in 6.15 seconds. The evidence supports FiberMargin as an auditable operator for irregular multiclass spatial fields, not as a universal mask smoother.

Keywords: multiclass spatial labels, categorical field repair, noisy labels, space-filling curves, selective prediction

1 Introduction

Semantic masks, anatomical label maps, spatial-omics domains, land-cover maps, and point-cloud annotations are all categorical fields on a spatial domain. Once such a field exists, a practitioner may trust its large-scale organization but suspect local assignment errors. Re-running the upstream annotator or segmentation model may be impossible, expensive, private, or unnecessarily disruptive.

We study *training-free multiclass spatial label repair*. The input is one label field $\tilde{y}_i \in \{1, \dots, K\}$ at coordinates $x_i \in \mathbb{R}^d$; the output is a repaired field and a pointwise audit map. Source images, expression values, logits, clean masks, and a training collection are unavailable. Regular pixel and voxel arrays are supported, while irregular 2D/3D point sets are the primary setting. This information budget separates repair from segmentation and from learning robustly with noisy annotations. The package also contains a separate binary utility, evaluated only as an auxiliary diagnostic in the supplement; it is not evidence for the multiclass enclosure mechanism developed here.

Section 2 situates this coordinate-and-label-only setting among classical spatial restoration, graphical models, learned segmentation refiners, noisy-label learning, annotation fusion, self-supervised denoising, and topology-aware segmentation. The supplement summarizes the information available to each method family.

FiberMargin represents each specimen through multiple one-dimensional charts made by rotating the coordinates and ordering them with a Hilbert space-filling curve (Sagan 1994; Moon et al. 2001). Each chart supports a linear-time recurrence. A single fixed transport range propagates unit class evidence in both path directions. The two directional amplitudes form an enclosure field that rewards bilateral support, while large coordinate jumps attenuate transport. The central decision rule is simple: accept a rival only when its mean rival-versus-observed contrast across the path atlas exceeds the barrier induced by chart disagreement, class prevalence, and local query isolation. The term *margin* refers to this difference between rival and observed-class evidence.

The paper makes four empirical and theoretical contributions:

1. an information-budget taxonomy for categorical-field cleaning, a formal training-free repair interface, and an exact identity separating repaired errors from damage to correct labels;
2. a two-sided enclosure margin based on an atlas of chain orderings, with an exact amplitude characterization, conditional chart-interior recovery results, and finite-input stability;
3. a reproducible benchmark spanning procedural multiclass masks, spatial simulations, volumetric fields, controlled biological coordinates, and two real upstream-prediction evaluations, with calibrated Potts, conditional-random-field (CRF), propagation, and multiscale-mode controls under the same information budget;
4. a minimal R API and C++ implementation measured through 500,000 observations and 20 classes.

2 Related Work and Task Positioning

2.1 Classical categorical-field restoration

Cleaning a spatial label field predates modern segmentation networks. Mathematical morphology provides erosion, dilation, opening, closing, reconstruction, and connected filtering driven by a chosen structuring element (Serra 1982; Soille 2003). Mode filters and connected-component rules similarly encode a local scale or an explicit size threshold. These operators are transparent and training free, but their behavior is tied to a lattice neighborhood and a fixed geometric template; applying the same template to a rare island, a narrow class, and an isolated error can produce opposite outcomes.

Conditional random fields (CRFs) are graphical models that combine a unary preference for each class with pairwise compatibility terms between locations (Geman and Geman 1984; Besag 1986). Iterated conditional modes is a fast local optimizer, while graph-cut constructions give strong optimization results for important discrete energies (Boykov, Veksler, and Zabih 2001). Fully connected CRFs add long-range pairwise interactions and typically combine positions with color or learned unary scores (Krähenbühl and Koltun 2011). CRF-RNN (*conditional random fields as recurrent neural networks*) unrolls CRF mean-field updates as recurrent layers, allowing them to be trained with a segmentation network (Zheng et al. 2015). Random-walker segmentation and harmonic label propagation spread information across an affinity graph (Grady 2006; Zhu, Ghahramani, and Lafferty 2003). FiberMargin is a coordinate-and-label-only restoration operator: it treats every observed label as potentially erroneous, rather than designating trusted seeds or using calibrated class probabilities. It aggregates spatial evidence along several rotated Hilbert paths, each evaluated by a one-dimensional recurrence. Under this same input constraint, we compare fixed Potts ICM and alpha-expansion Potts inference, a coordinate-only CRF, self-anchored harmonic propagation, and multiscale mode filtering. Methods that also use image features, calibrated unaries, or trusted seeds operate in richer input settings and are discussed separately.

2.2 Feature-guided and learned mask refinement

A large computer-vision literature improves an existing segmentation by returning to the source image or its learned representation. GrabCut couples user initialization, color models, and graph-cut optimization (Rother, Kolmogorov, and Blake 2004). PointRend adaptively resamples uncertain points using backbone features (Kirillov et al. 2020). CascadePSP refines a coarse mask from global and local image context (H. K. Cheng et al. 2020), SegFix learns directions from boundary pixels toward reliable interiors (Yuan et al. 2020), and Boundary Patch Refinement retrains high-resolution image-and-mask patches around predicted contours (Tang et al. 2021). Mask Transfuser detects incoherent quadtree regions and updates them with learned image features (Ke et al. 2022). These methods principally target contour fidelity or high-resolution instance masks.

More recent systems learn a generic repair prior rather than attaching refinement to one upstream architecture. SegRefiner learns a conditional discrete diffusion process from coarse/clean mask pairs (Wang et al. 2023), while RNCA learns recurrent self-repair using image context and reference masks (Silbernagel et al. 2026). Both learn from paired masks, and RNCA additionally uses image context. They address a training-enabled setting that

is distinct from the single-field repair task studied here. Our external-mask experiments test the stricter mask-only operator, not superiority over feature-guided refiners.

2.3 Noisy-label learning, dataset cleaning, and blind denoising

Learning with noisy labels usually asks how to estimate a predictor from many feature-label examples despite corruption. Representative strategies estimate a noise transition and correct the loss (Patrini et al. 2017), exchange small-loss examples between networks (Han et al. 2018), infer likely dataset label errors from out-of-sample class probabilities (Northcutt, Jiang, and Chuang 2021), or use instance-level confidence information (Berthon et al. 2021). Their output is a robust predictor, a ranked error list, or a revised training set. FiberMargin does not train a predictor. It revises one supplied categorical field once, using only its coordinates and observed labels.

Pixelwise annotation noise has also motivated segmentation-specific training methods. ADELE uses early network predictions to update noisy masks during training (Liu et al. 2022); Spatial Correction models correlated annotation errors and progressively corrects them (Yao et al. 2023); and PNAL combines prediction history with clusterwise correction for noisy 3D point-cloud training (Ye et al. 2021). These methods are highly relevant to the application but require images or point features, a trainable segmentation model, and repeated optimization. They address *learning from noisy masks*, whereas our task is *repairing a supplied mask when the learning system is unavailable*.

Noise2Self provides a closer conceptual analogy: leave-out prediction can denoise measurements without clean targets when noise is conditionally independent across measurement dimensions (Batson and Royer 2019). FiberMargin’s leave-self-out transport likewise prevents a query label from certifying itself. The objectives nevertheless differ. Noise2Self estimates or trains a denoising function for numerical measurements under a $\mathcal{J}$ -invariance argument; FiberMargin makes categorical edits from spatial redundancy and does not estimate deployment risk from the noisy field alone.

2.4 Repeated annotations, topology, and selective action

When several labels of the same object are available, aggregation is statistically better identified than one-mask repair. Dawid-Skene estimates annotator confusion rates (Dawid and Skene 1979), and STAPLE jointly estimates a latent segmentation and the performance of multiple raters or algorithms (Warfield, Zou, and Wells 2004). Multi-annotator segmentation models extend this idea using images and learned annotator models (Tanno et al. 2019; Zhang et al. 2023). FiberMargin takes one label field as input, so it cannot compare annotators or estimate how reliable each one is. When several annotations of the same object are available, they should be combined with an annotation-fusion method or studied in a separate multi-annotation analysis.

Topology-aware segmentation addresses a different failure mode. Persistent-homology losses constrain Betti structure during network training (X. Hu et al. 2019), and cDice targets connectivity of tubular structures through centerlines (Shit et al. 2021). These methods use clean topology-bearing references or an explicit structural target. FiberMargin has neither. Its adaptive barrier is instead related to selective prediction: it acts only when evidence clears a threshold (El-Yaniv and Wiener 2010), but the deterministic score is not

a calibrated probability or a topology certificate. We therefore report Boundary IoU (B. Cheng et al. 2021), rare-class measures, correction recall, and damage, and retain the thin-network failure as a counterexample.

2.5 Spatial omics and the exact comparison scope

Spatial transcriptomics is the principal irregular-domain application here. SpaGCN, GraphST, BayesSpace, BANKSY, and KODAMA estimate spatial domains from gene expression and spatial coordinates, with histology incorporated by some methods (J. Hu et al. 2021; Long et al. 2023; Zhao et al. 2021; Singhal et al. 2024; Abdel-Shafy et al. 2025). FiberMargin starts only after a categorical assignment exists and never claims to rediscover domains. We compare the optional post-hoc correction rules from SpaGCN and GraphST, not their clustering models, because those rules accept labels and coordinates. The same distinction applies beyond transcriptomics: a point-cloud network trained under noisy labels and a post-hoc repair of its exported semantic field are complementary stages.

Across these literatures, comparability is governed by six questions: Is there one mask or repeated annotations? Are source measurements, image features, or logits available? Are any labels trusted? Is a training collection available? Is topology specified? Must the method work natively on irregular 2D/3D coordinates? FiberMargin occupies the restrictive cell with one fallible field, coordinates only, no fitted corpus, no anchors, and no topology template. The contribution is a new decision operator for this cell, not a claim to have invented Hilbert orderings, exponential attenuation, relative margins, or selective decisions (Rockafellar and Uryasev 2000; El-Yaniv and Wiener 2010). Its defining multi-class construction is the two-sided path enclosure: opposite directional states are combined before class normalization, and a rival must remain positive across distinct spatial flattenings strongly enough to clear their observed disagreement. The Supplement compares this operator with additive transport, one reference path, smaller atlases, and reduced admission barriers.

3 FiberMargin

3.1 Mask-repair objective and specimen isolation

Let y_i^* denote an evaluation-only reference, $\tilde{y}_i$ the imperfect input, and $\hat{y}_i$ the repaired output. Define initial and repaired accuracy as A_0 and A_1 , correction recall $R = \Pr(\hat{y} = y^* \mid \tilde{y} \neq y^*)$, and damage $D = \Pr(\hat{y} \neq y^* \mid \tilde{y} = y^*)$. Partitioning cells by initial correctness gives the exact identity

$$A_1 = A_0(1 - D) + (1 - A_0)R, \quad A_1 - A_0 = (1 - A_0)R - A_0D.$$

Thus mask cleaning is not synonymous with smoothness: an edit policy improves accuracy only when corrected error mass exceeds damaged correct mass. We report both terms, together with class-balanced and boundary measures, rather than selecting a method by accuracy alone. At deployment y^* is absent; FiberMargin therefore returns a candidate, its uncalibrated `margin_score`, the required barrier, `repair_margin = margin_score - required`, and the final retain/change decision. These diagnostics are deterministic evidence contrasts, not probabilities or estimates of expected loss.

Let $s_i \in \{1, \dots, S\}$ identify a specimen or independent mask. FiberMargin applies the complete operator separately to each set $I_s = \{i : s_i = s\}$. Coordinates, the locally observed class alphabet, class counts, path states, and thresholds never cross specimens. This is stronger than assigning zero weight to distant sections: the output and every pointwise diagnostic for one specimen are mathematically invariant to all coordinates and labels in every other specimen, including specimens that use the same numeric coordinate range.

Within specimen s , axes with zero empirical range are removed before normalization. Each remaining coordinate dimension is mapped by its empirical 1st and 99th percentiles to a robust unit interval and clipped to $[-0.1, 1.1]$. We write the normalized coordinate as $z_i \in \mathbb{R}^{d_s}$, where $d_s \in \{2, 3\}$; inputs with fewer than two varying axes in any specimen are rejected. Thus a constant third coordinate invokes exactly the 2D operator, whereas a varying third coordinate participates in rotations, Hilbert ordering, path distances, and the binary distance calculation.

3.2 Path atlas and two-sided enclosure

For view a , a fixed rotation R_a is applied to z_i . Each rotated coordinate axis is rescaled to $[0, 1]$ for a 16-bit Hilbert code, yielding a permutation $\pi_a = (\pi_{a1}, \dots, \pi_{an})$. The atlas contains nine paths in 2D and 12 in 3D. Each path is determined solely by its rotation and Hilbert ordering.

Let $\Delta_{at} = \|z_{\pi_{at}} - z_{\pi_{a,t-1}}\|_2$ and let r_a be the median positive path step. At an interior query $i = \pi_a(t)$ define $q_{ai} = \sqrt{\Delta_{a,t-1}\Delta_{at}}$; at an endpoint use the available step. The transport action and query-isolation factor are

$$h_{at} = \frac{\Delta_{at}}{r_a}, \quad \omega_{ai} = \max \left\{ 1, \frac{q_{ai}}{r_a} \right\}.$$

For class indicator $z_{ajc} = \mathbf{1}\{\tilde{y}_j = c\}$, one forward and one reverse recurrence at the fixed range $\lambda = 5$ give

$$L_{aic} = \sum_{j \prec_a i} e^{-D_a^+(i,j)/5} z_{ajc}, \quad R_{aic} = \sum_{j \succ_a i} e^{-D_a^-(i,j)/5} z_{ajc},$$

where D_a^+ and D_a^- are sums of the directed actions. The strict path inequalities are leave-self-out: a query's supplied label is inserted only after its directional state has been read. FiberMargin combines directional amplitudes as

$$H_{aic} = (\sqrt{L_{aic}} + \sqrt{R_{aic}})^2, \quad p_{aic} = \frac{H_{aic}}{\max\{\sum_r H_{air}, 10^{-12}\}}.$$

The cross term rewards evidence that encloses a query from both path directions while retaining unilateral evidence. Sources have unit mass; local isolation affects only the final acceptance decision.

3.3 Atlas margin and selective decision

Relative to the supplied class $\tilde{y}_i$, define the path contrast $m_{aic} = p_{aic} - p_{ai, \tilde{y}_i}$. For A paths, the atlas margin and descriptive chart-disagreement scale are

$$m_{ic} = \frac{1}{A} \sum_a m_{aic}, \quad s_{ic} = \frac{1}{A} \left\{ \sum_a (m_{aic} - m_{ic})^2 \right\}^{1/2}.$$

The second expression is the empirical population standard deviation across the deterministic paths divided by $\sqrt{A}$; it is not a calibrated confidence interval. The candidate $c_i^* = \arg \max_c m_{ic}$ is initialized at the supplied class, so exact ties retain that class. With observed class count n_c , median positive class count n_{med} , and median isolation $\omega_i = \text{median}_a \omega_{ai}$,

$$\delta_c = \sqrt{\frac{n_{\text{med}}}{\max(n_c, 1)}}, \quad B_{ic} = \frac{s_{ic}\omega_i}{\delta_c}.$$

FiberMargin changes $\tilde{y}_i$ to c_i^* only when $c_i^* \neq \tilde{y}_i$ and $m_{ic_i^*} \geq B_{ic_i^*}$. The returned `margin_score` is $m_{ic_i^*}$, `required` is $B_{ic_i^*}$, and `repair_margin` is their difference. These are uncalibrated deterministic contrasts, not probabilities.

For exactly two observed classes, the implementation uses one strict local ballot over $\min(n-1, 2^{d+4})$ nearest other locations; a tie retains the supplied class and a distinct candidate requires one net vote. Specimens with fewer than 20 locations, fewer than two observed classes, or no coordinate extent are returned unchanged. In a binary field the rival is unique, so the signed difference between local rival and supplied-class counts provides the complete local candidate comparison without multiclass path normalization. This binary specialization is fixed and is not user-selectable.

3.4 Characterization and conditional guarantees

The square-root enclosure has a direct characterization. Within fields of the form $F(L, R) = (g(L) + g(R))^2$ with nonnegative g and $g(0) = 0$, exact retention of one-sided evidence, $F(L, 0) = L$ and $F(0, R) = R$, forces $g(u) = \sqrt{u}$ and therefore $F(L, R) = (\sqrt{L} + \sqrt{R})^2$. This applies to the unnormalized field and does not derive the atlas size, rotations, or range.

The Supplement proves deterministic path-weight, finite-input recurrence-stability, and specimen-isolation results for the implemented recurrences. Its binary and multiclass recovery theorems are conditional chart-interior results under stated witness and label-noise assumptions, not universal consistency claims. In particular, they do not imply recovery of coherent regional overwrites or correctness at every boundary.

4 Experimental Design

The evaluation separates six questions that a single headline accuracy cannot answer: whether the operator corrects known errors under controlled geometry and density; whether it transfers to unseen shapes and native 3D coordinates; whether it works on real tissue arrangements without expression or images; whether a regular raster instead favors a grid-specific smoother; which parts of the operator matter; and whether the C++ implementation remains practical at large sample sizes. In every corruption study, the reference field is retained only for scoring. It is never an input to FiberMargin or a comparator.

4.1 Evaluation units

For simulation inference, the independent unit is a geometry family rather than one corruption draw: repeats on the same field share its class layout, boundary geometry, and sampling pattern. Confidence intervals therefore use 10,000 paired bootstrap resamples of whole-family means. Real-coordinate perturbations are descriptive recovery tests on a fixed tissue arrangement, not biological replicates.

For the genuine upstream-mask evaluation, an independently annotated CamVid test frame is the summary unit. Frames retain sequence dependence, so image-wise summaries are descriptive rather than formal out-of-domain significance tests.

4.2 Controlled spatial recovery studies

Purpose. Controlled fields make the correct class known at every location, so they can measure both repaired errors and newly damaged correct labels while varying one spatial difficulty at a time. The complete simulation suite contains 708 cases from 12 planar or layered geometry families, including jagged and curved stripes, thin layers, branching regions, rings, disconnected components, density gradients, 2D/3D A-B-C gradients, five region-concentration profiles, four corruption mechanisms, and one or three specimens. It asks whether recovery persists across the forms of class shape, sampling imbalance, and local error that occur in spatial annotations. A separate 108-case unseen-shape suite uses six held-out geometry families; it tests transfer to new formulas rather than another draw from the primary shapes. Every procedural corruption retains one correct exemplar of each true class represented in a specimen. The target is repair of labels that remain represented in the supplied field, rather than discovery of a class absent from that field. Figure 1 compares Accuracy and ARI across both simulation ledgers and the dedicated variable-density 3D panel.

The separate, non-overlapping volumetric suite asks whether the same operator works on genuinely three-dimensional arrangements rather than stacked planar examples. It crosses eight 3D structures with uniform, class-imbalanced, or irregular-plane acquisition and random, boundary, patch, or regional corruption. It contains 288 cases of 12,000 points and 32 size-confirmation cases of 60,000 points.

Controlled biological-coordinate studies ask the same question on real spatial arrangements while keeping recovery measurable: annotations are first treated as the reference and then locally corrupted. This design does not claim correction of unadjudicated biological annotation errors. We use 47,329 spots from 12 DLPFC sections and three donors (Maynard et al. 2021; Pardo et al. 2022) to test layered, multi-section tissue; 28,317 cells from five consecutive MERFISH hypothalamic planes from one mouse (Moffitt et al. 2018; Li and Zhou 2022) to test a registered multi-plane arrangement; and 194,541 annotated locations from a 19-class colorectal cancer (CRC) Visium HD tissue derived from the 10x Genomics *Visium HD Spatial Gene Expression Library, Human Colorectal Cancer (FFPE)* dataset, analyzed with Space Ranger 4.0.1 (10x Genomics 2025), to test a dense, multiclass tissue field. The CRC WSI region classes were derived by the authors. Labels are replaced by labels from plausible adjacent regions under fixed random, interface, compact-patch, and regional protocols. FiberMargin receives only the resulting coordinates, corrupted labels, and specimen identifiers.

4.3 Procedural raster-mask diagnostic

This diagnostic asks a deliberately difficult question for a general coordinate-based operator: can it repair a regular categorical raster when no source image or model is available, and how much does a fixed grid-local smoothing prior help? The suite crosses six 96x96 families—nested regions, jagged layers, thin networks, disconnected islands, touching regions, and perforated rings—with three geometry seeds, independent, boundary, or coherent-patch errors, and 15% or 30% corruption, for 108 cases. The families respectively stress containment, irregular boundaries, narrow connectivity, rare components, close class contacts, and holes. Every method receives only the corrupted categorical array. The same exported package functions create corruptions, clean masks, and score outputs.

4.4 Auxiliary binary building-mask diagnostic

This is a controlled cross-domain geometry test. It asks whether the unchanged mask-only operator can repair held-out non-transcriptomic mask geometry, while controlled corruptions make the exact recovery and damage observable. We used 55 256x256 masks from a held-out project-separated test split of the HOT Very-High-Resolution Building Segmentation Dataset (Humanitarian OpenStreetMap Team 2026). Eligibility required at least 256 pixels of each class. Only binary masks were read, never the aerial images. Each source mask received random, boundary-concentrated, and compact-patch corruption at 10% and 25%, producing 330 cases. This tests recovery from injected defects, not segmentation from imagery or correction of naturally occurring mapping errors.

Because every HOT field is binary, this study evaluates the package’s fixed local ballot utility rather than the multiclass path-enclosure operator. Its results are reported only in the supplement as a package-scope diagnostic and are not used as evidence for the enclosure mechanism.

The matched comparison contains nine coordinate-and-label rules: FiberMargin, fixed Potts ICM, alpha-expansion optimization of a fixed Potts energy, a coordinate-only CRF, self-anchored harmonic propagation, multiscale mode filtering, fixed SpaGCN correction and GraphST correction rules, and a one-pass 8-NN modal filter. Every rule receives the same coordinates, observed categorical field, and specimen identifier in every study. Results average the six corruption conditions within each source mask, which is the independent evaluation unit.

4.5 Controlled public-annotation diagnostics

To test ordinary segmentation-mask geometry without turning the study into an image-refinement benchmark, we used the categorical annotations of 24 Oxford-IIIT Pet images (Parkhi et al. 2012) and 24 CamVid images (Brostow, Fauqueur, and Cipolla 2009). Oxford trimaps were converted to foreground/background and CamVid retained its annotated semantic classes, with void pixels excluded. Each mask was nearest-neighbor resized to a maximum side length of 128, then subjected to random, boundary, and compact-patch errors at 15% and 30%, for 144 corruptions per collection. RGB imagery was never read. These are controlled annotation-corruption diagnostics, not evaluations of naturally predicted masks.

4.6 Real upstream-mask evaluations

We separately evaluate repair of unmodified upstream categorical predictions against independent references. For CamVid (Brostow, Fauqueur, and Cipolla 2009), a fixed Extremely Randomized Trees semantic pixel classifier (Geurts, Ernst, and Wehenkel 2006) was fitted only on the official training frames using RGB, locally averaged RGB, edge magnitude, and normalized pixel position. It then predicted all 233 official test frames; the 11 retained semantic classes were evaluated at a maximum side length of 160 with void reference pixels excluded. The test annotations were never used to fit the upstream classifier.

For S3DIS (Armeni et al. 2016), the same classifier family was fitted to Areas 1, 2, 3, 4, and 6 using XYZ and RGB, then evaluated on one deterministic 4,096-point block from each of 68 Area 5 rooms. FiberMargin and the controls received only XYZ and the saved categorical prediction. These deliberately simple upstream models create independent prediction/reference pairs for testing the repair interface; they are not intended as state-of-the-art segmentation benchmarks.

Images and point features are used only to produce the upstream fields. FiberMargin and every equal-input control receive only a categorical prediction and coordinates, never RGB values, detector scores, logits, or reference labels.

4.7 Fixed comparator calibration

The four calibrated controls were set before the reported evaluations. A non-overlapping development panel of 24 simulated 2D/3D fields used distinct geometry formulae, corruption draws, and random seeds from the reported suites. For each control family, we selected the setting with the largest mean accuracy subject to the same absolute damage cap of 0.05; ARI broke accuracy ties. SpaGCN and GraphST retain their fixed published correction settings, and the modal reference retains $k = 8$. No comparator setting was selected from a reported test result. The calibration files include every candidate setting and case-level result.

4.8 Comparators and metrics

The direct comparison asks whether a rule improves the same supplied categorical field under the same information budget. Numerical eligibility therefore requires a post-hoc categorical rule that consumes exactly one categorical field, coordinates, and specimen identifiers, without images, expression, logits, trusted anchors, repeated annotations, or a fitted training collection. Let Y be the one-hot encoding of the observed labels and let P_k be a row-normalized k -nearest-neighbor coordinate matrix constructed separately in each specimen. The four calibrated control families use only these inputs.

The first family is a fixed local Potts-style ICM rule, and the second is standard multi-label alpha-expansion graph-cut inference (Boykov, Veksler, and Zabih 2001) for

$$E(z) = 5 \sum_i \mathbf{1}\{z_i \neq \tilde{y}_i\} + \sum_{\{i,j\} \in G_8} \mathbf{1}\{z_i \neq z_j\},$$

where G_8 is the undirected, symmetrized eight-nearest-neighbor graph built within each specimen or mask. Alpha expansion starts from the observed field and performs three com-

plete class cycles. Its graph construction, unary weight, pairwise weight, and cycle count are fixed across evaluations; the ICM rule is a separately configured eight-pass local optimizer. These controls distinguish global discrete-energy optimization from local smoothness updates.

The coordinate-only CRF is a sparse Gaussian-coordinate conditional random field evaluated by eight mean-field updates,

$$Q_i^{(t+1)}(c) \propto \exp \left[-1.5 \mathbf{1}\{c \neq \tilde{y}_i\} + 2.5(P_{16}Q^{(t)})_{ic} \right].$$

It uses no image features, logits, learned unaries, or reference labels. The self-anchored harmonic-propagation control supplies every observed label as an equally weighted soft observation, rather than designating a trusted seed set:

$$F^{(t+1)} = 0.20Y + 0.80P_{24}F^{(t)}, \quad t = 0, \dots, 11.$$

Finally, the multiscale-mode control averages categorical support over $k \in \{6, 14, 30\}$ coordinate neighborhoods, adds 0.20 support to the observed class, and changes a label only when the best rival exceeds its observed-class support by 0.05. Thus it is a multiscale version of the mode-filter family, defined on the same irregular 2D/3D coordinates as FiberMargin rather than only on a pixel lattice. Alpha expansion, the coordinate-only CRF, self-anchored propagation, multiscale mode, and Potts ICM are joined by the fixed SpaGCN correction rule, fixed GraphST correction rule, and the one-pass 8-NN modal reference. These nine rules are evaluated on every controlled and real-upstream benchmark: planar and volumetric simulations, CRC, DLPFC, MERFISH, procedural masks, held-out building masks, and image masks. We compare the coordinate-label correction rules rather than end-to-end clustering systems. The two published-rule reproductions were checked pointwise against official source commits (SpaGCN developers 2026; GraphST developers 2026). They matched all 9,600 predictions on continuous coordinates; 5 of 648 lattice predictions differed because exact-distance ties have no specified stable order.

The modal reference is a one-pass instance of classical local mode filtering (Serra 1982; Soille 2003). For each labelled location i , let $\mathcal{N}_k(i)$ be its k nearest non-focal labelled locations within the same specimen or mask, and let $\tilde{y}$ denote the corrupted labels. The filter returns

$$M_k(i) \in \arg \max_c \sum_{j \in \mathcal{N}_k(i)} \mathbf{1}\{\tilde{y}_j = c\}.$$

The implementation resolves a tied mode with the nearest tied neighbour. The compact $k = 8$ rule is part of the matched nine-method roster. A broad $k = 48$ version is a separate regular-grid diagnostic, shown only for raster-mask studies; neither value is tuned per mask or corruption condition. ADELE and PNAL require a trainable predictor, data features, and prediction history; Noise2Self requires a numerical self-supervision setting; image-guided refiners require image features or logits; and annotator aggregation requires repeated labels. These methods are relevant neighboring tasks, but their information contracts differ. Supplement S1 gives the full input taxonomy.

No single score captures useful repair. The evaluation maintains four core measures: accuracy and ARI measure global recovery, correction recall measures recovery among initially wrong labels, and damage measures mistakes introduced among initially correct labels.

Each main-text table shows the subset most relevant to its question, while the supplement reports the complete core outcomes and secondary diagnostic measures: mean IoU for mask overlap; one-grid-step Boundary IoU (B. Cheng et al. 2021) for geometric edge recovery; macro and worst recall for class balance; and boundary and rare/sparse-region accuracy for difficult locations. For a reference field y^* and evaluation neighborhood $\mathcal{N}_i$, the generic boundary indicator is $b_i = \mathbf{1}\{\exists j \in \mathcal{N}_i : y_j^* \neq y_i^*\}$, with anatomical class-pair restrictions for DLPFC and MERFISH. If $U(i)$ is an evaluation unit and r_i a reference region or class, let $n_{Uc} = \sum_{i:U(i)=U} \mathbf{1}\{r_i = c\}$, $c_U^- = \arg \min_{c:n_{Uc}>0} n_{Uc}$, and $s_i = \mathbf{1}\{r_i = c_{U(i)}^-\}$; no prevalence-percentage cutoff is used. The neighborhood sizes, the 20% geometric-boundary threshold used by simulations, tie handling, and all dataset-specific exceptions are fixed in Supplement S4. Correction recall and damage are the two conditional terms in the exact identity above. The shared core metrics make the correction–protection trade-off visible in every main-text comparison.

The scaling study is an engineering test of the complete R-to-C++ call rather than a claim of a universal runtime ranking. Supplement S3 reports matched one-worker timing. A timed call includes validation, specimen splitting, coordinate normalization, geometry construction, inference, and output assembly; it excludes file I/O, corruption generation, reference-derived masks, scoring, and plotting. Hardware, compiler flags, and separate deterministic CPU-parallel verification are reported there.

5 Results

5.1 Broad multiclass controlled recovery

Question. Does the two-sided enclosure margin improve multiclass labels across prespecified and previously unseen spatial structures? The complete suite varies geometry, density, class balance, dimensionality, and corruption mechanism; the unseen suite introduces new geometry formulas; and the 3D panel tests native variable-density volumes. Figure 1 summarizes 36 variable-density 3D fields, 708 complete simulations, and 108 unseen geometries. Every method receives the same coordinates and corrupted categorical labels.

FiberMargin obtained the highest Accuracy and ARI on both the complete and unseen-geometry ledgers. It also obtained the highest Accuracy on variable-density 3D, while multiscale mode had a 0.0010 higher ARI. Descriptive paired family-clustered intervals compare FiberMargin with the strongest displayed control; they are not selection-adjusted. The complete-suite Accuracy difference was 0.0112 (95% interval 0.0076–0.0146), and the unseen-suite difference was 0.0108 (0.0078–0.0147). In the variable-density 3D panel, the Accuracy interval crossed zero and the ARI difference was compatible with a tie. Thus the strongest evidence supports broad multiclass recovery across irregular geometries, not a universal ranking over every spatial repair problem.

5.2 Controlled biological-coordinate recovery

This section reports three real-coordinate studies. **Question.** Given the actual coordinate layout of layered tissue, consecutive planes, or a dense 19-class field, can a method repair deliberately introduced local label errors without using expression data, images, or the reference annotation? The published or author-derived labels are treated as evaluation-only

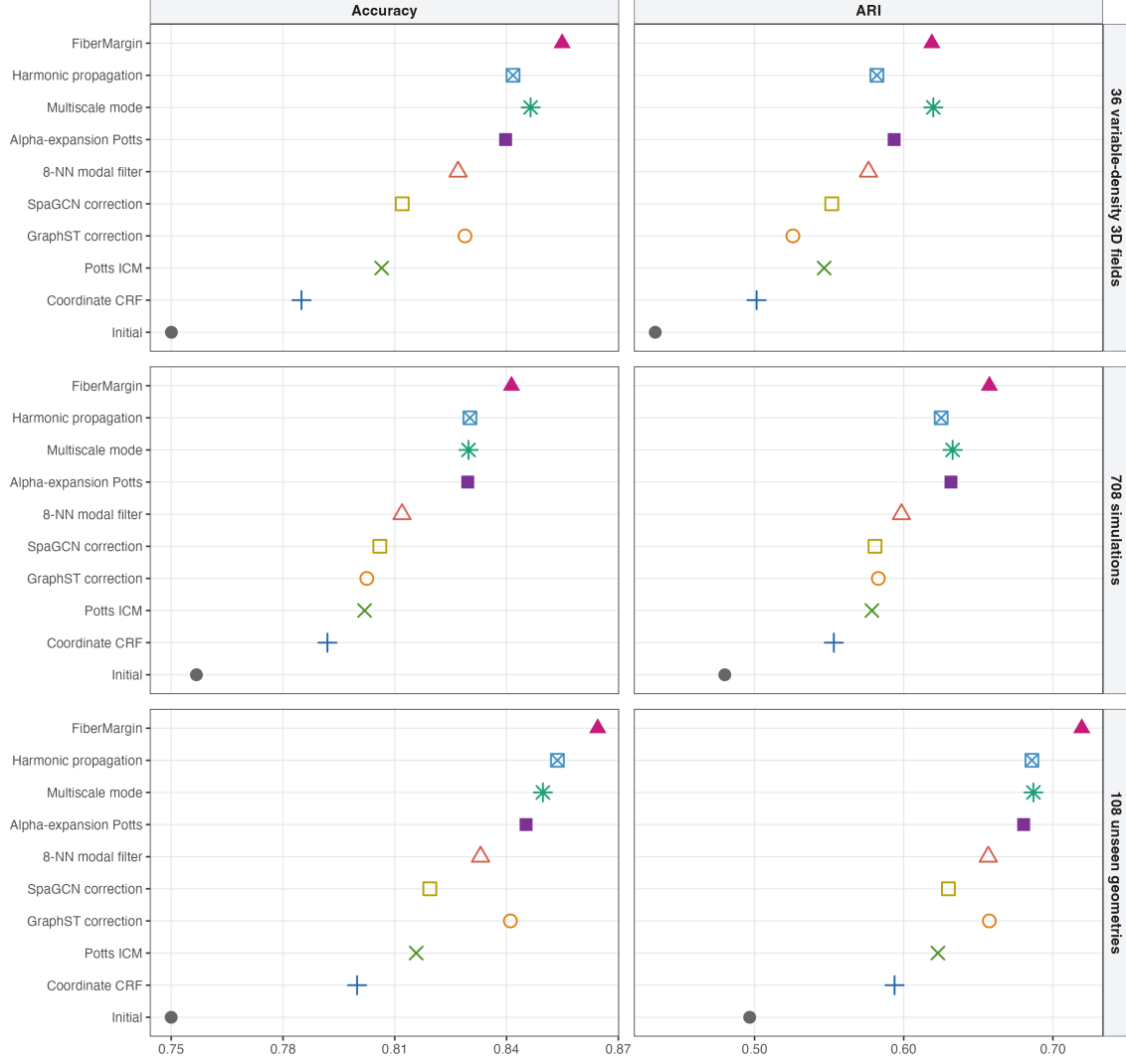


Figure 1: Accuracy and adjusted Rand index across 36 variable-density 3D fields, 708 complete simulations, and 108 unseen geometries. These are evaluations of FiberMargin’s multiclass two-sided enclosure mechanism. Every method receives the same coordinates and corrupted categorical labels; color and shape jointly identify the method.

references, fixed neighbor-plausible corruptions are introduced, and the original labels are hidden before inference. These studies therefore establish controlled recovery on biological geometry, not correction of naturally disputed annotations.

FiberMargin had the highest aggregate Accuracy and ARI on DLPFC (0.8942 / 0.8220) and MERFISH (0.8905 / 0.8004) among the nine matched refiners. CRC was the exception: multiscale mode narrowly led FiberMargin (0.8604 / 0.7827 versus 0.8589 / 0.7806). FiberMargin recovered a larger fraction of corrupted CRC labels (0.4444 versus 0.4105) but incurred more damage (0.0478 versus 0.0336), making this a clear operating-point trade-off rather than a uniform win. Detailed DLPFC and MERFISH operating points, boundary Accuracy, sparse-region Accuracy, and class-balance comparisons are reported in the supplement.

The predefined random 25% CRC visualization illustrates what aggregate means can and cannot show (Figure 2). FiberMargin raised Accuracy from 0.7500 to 0.9715 and corrected 0.9204 of injected errors with damage 0.0114. In contrast, one 25% coherent-patch case fell from 0.7500 to 0.7104. A contiguous patch overwritten with a neighboring class is observationally ambiguous when only coordinates and the categorical field are available; the failure is retained in the scenario archive.

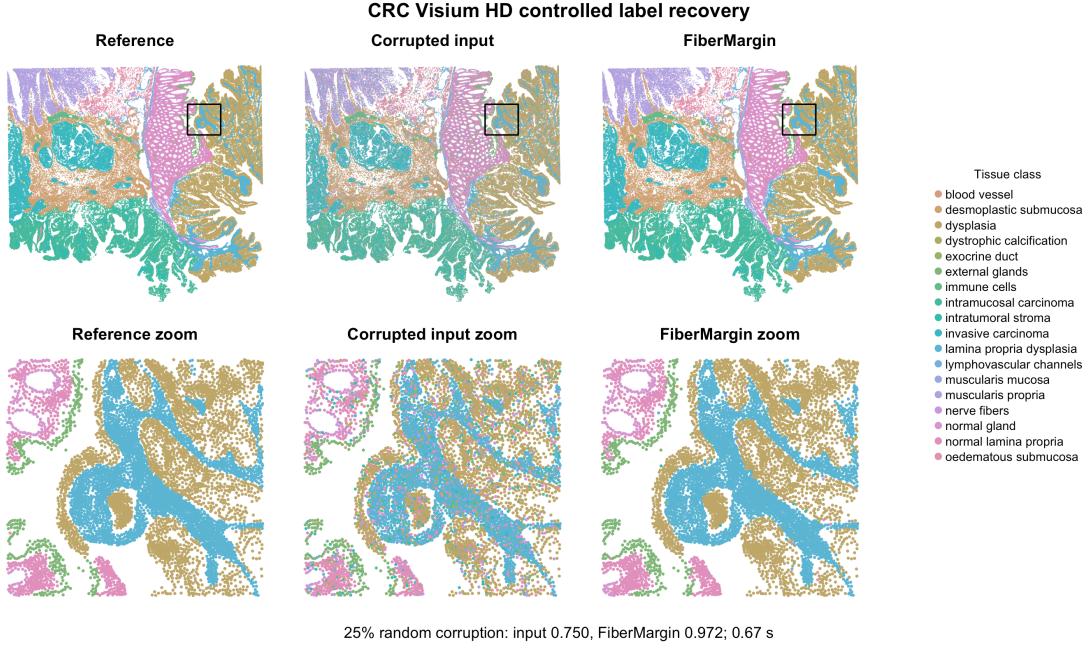


Figure 2: Full-tissue and zoomed 19-class CRC Visium HD recovery under 25% random corruption. FiberMargin uses its multiclass two-sided enclosure mechanism. The black rectangle identifies the zoom.

5.3 Procedural multiclass mask repair

Question. Does the same multiclass enclosure remain useful on regular categorical rasters, where grid-specific local filters have a natural advantage? The suite crosses six 96-by-96 geometry families, three error mechanisms, two corruption levels, and three seeds, producing 108 cases. Table 1 includes the complete fixed comparator roster plus a separate 48-NN regular-grid diagnostic.

The 48-NN filter had the highest control Accuracy (0.9070), while the 8-NN filter had the highest control ARI and correction recall. FiberMargin reached 0.9113 Accuracy and 0.7392 ARI, leading both primary scores, and corrected 0.6210 of altered locations. Its 0.0056 damage was higher than multiscale mode (0.0011) and alpha expansion (0.0009), so the result is an Accuracy–damage trade-off rather than a free gain from changing more cells.

Table 1: Aggregate core metrics on 108 procedural multiclass mask-repair cases. Higher is better except for damage. Every matched comparator receives the same corrupted categorical field and coordinates; the 48-NN modal filter is an additional regular-grid diagnostic. Secondary overlap, boundary, and rare-class metrics are reported in the supplement.

Method	Accuracy	ARI	Correction	Damage
Initial	0.7750	0.4242	0	0
SpaGCN correction	0.8740	0.6335	0.4573	0.0018
GraphST correction	0.9059	0.6964	0.6362	0.0163
8-NN modal filter	0.9023	0.7218	0.6456	0.0231
48-NN modal filter	0.9070	0.7012	0.6379	0.0153
Alpha-expansion Potts	0.8997	0.7037	0.5511	0.0009
Coordinate CRF	0.8662	0.6174	0.4166	0.0001
Harmonic propagation	0.8961	0.6913	0.5357	0.0019
Multiscale mode	0.9043	0.7210	0.5781	0.0011
Potts ICM	0.8988	0.7009	0.5498	0.0017
FiberMargin	0.9113	0.7392	0.6210	0.0056

The operating-point audit multiplied the fixed barrier by two, retained the default, or removed the barrier. Correction recall and damage were respectively 0.584/0.0021, 0.621/0.0056, and 0.645/0.0149. Qualitatively, FiberMargin cleaned isolated and disconnected regions but removed much of one very thin network. That counterexample, retained in the supplement, prevents a topology-preservation claim.

5.4 Operator diagnostics

These diagnostics test the final operator rather than add method comparisons. The supplement reports the complete atlas-size audit with paired Accuracy, ARI, worst-recall, sparse-region, and damage outcomes. The results support the fixed 9-path 2D and 12-path 3D atlas used by the package; changing CPU workers does not change the mathematical operator.

The specimen-isolation test verifies the stated mathematical contract rather than recovery Accuracy: translating, rotating, coordinate-permuting, or label-permuting a distractor specimen changed zero of 2,600 target predictions. This exact invariance follows from independent specimen processing.

The software contract additionally tests exact equality of all pointwise diagnostics under one, two, and four requested CPU workers; row-permutation equivariance; translation and

uniform-scale invariance; equality of 2D and constant- z inputs; and non-equivalence after removing informative z from a genuinely 3D field. A mixed call may therefore contain both effective 2D and effective 3D specimens without sharing coordinate normalization or class evidence.

5.5 Scaling

This feasibility study measures the complete R-to-C++ call rather than a universal runtime ranking. Fresh R processes measured complete 20-class calls at 10,000, 50,000, 100,000, 250,000, and 500,000 observations. Runtime rose from 0.067 to 6.146 seconds. The fitted log-log slope was 1.166 ($R^2 = 0.998$). At 500,000 observations, peak resident memory was 569 MB and the increase during refinement was 422 MB. For fixed atlas size, transport range, and class count, time is $O(n \log n + nK)$ and stored chart actions require $O(nK|\mathcal{A}|)$ memory. Matched cross-method timings are reported in Supplement S15.

5.6 External transfer and auxiliary scope tests

The final evaluations ask whether the operator transfers beyond controlled recovery to categorical masks produced by real upstream predictors. All repair methods receive the same saved categorical prediction and coordinates; source features and upstream outputs beyond the categorical prediction are withheld. Figure 3 shows the matched Accuracy–damage operating points, and the supplement reports the complete image- and room-level matrices.

FiberMargin improved CamVid Accuracy by 0.0186 (descriptive 95% image-bootstrap interval 0.0175–0.0197) and S3DIS Accuracy by 0.0204 (room-block interval 0.0152–0.0259). GraphST correction had the highest Accuracy in both studies; its advantage over FiberMargin was 0.0061 on CamVid and 0.0112 on S3DIS. Harmonic propagation also provided a more favorable Accuracy–damage point in these two settings. Frames and room blocks retain sequence or scene dependence, so the intervals are descriptive. These tests establish that FiberMargin can improve saved predictions, but they also delimit its current transfer range: the strongest evidence remains irregular multiclass controlled recovery.

The supplement separately reports 330 controlled corruptions of 55 held-out binary building masks. That study evaluates the package’s local-ballot utility, not the multiclass enclosure developed in this paper, and is therefore excluded from the primary evidence hierarchy.

6 Discussion

FiberMargin addresses a deliberately restricted multiclass repair task with one explicit margin-generating recurrence. Each remote observation contributes through a known exponential path weight, the fixed path atlas determines the contrast, and a positive gap makes the decision stable. The C++ implementation retains this structure closely enough for source-level audit.

The experiments identify a coherent regime of advantage. FiberMargin led the complete and unseen-geometry simulation ledgers, procedural multiclass masks, DLPFC, and MER-FISH, and attained the highest Accuracy in the variable-density 3D study. These settings

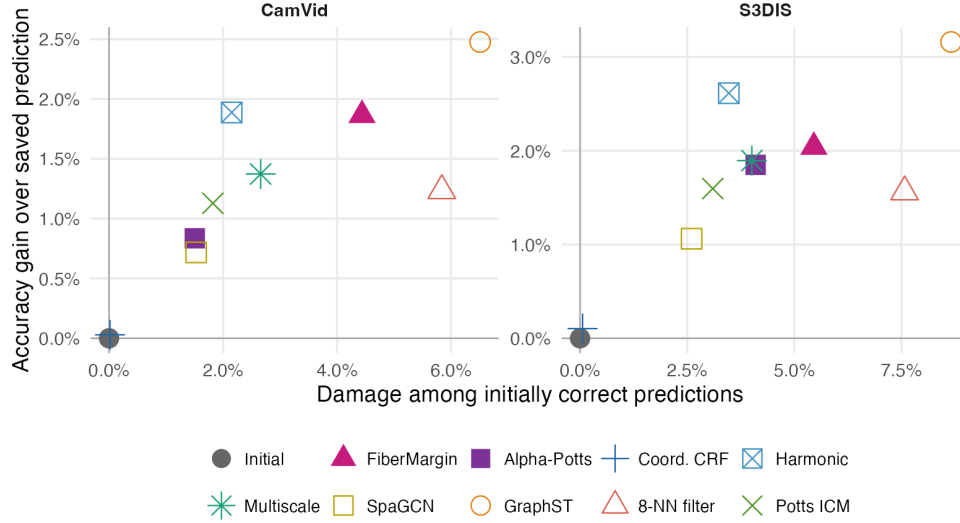


Figure 3: Accuracy gain and correction damage for saved CamVid and S3DIS upstream predictions. Both evaluations are multiclass and use FiberMargin’s two-sided enclosure mechanism. Every repair method receives only the categorical prediction and coordinates; color and shape jointly identify methods.

share multiclass structure, irregular or heterogeneous spatial support, and errors that remain locally contradicted from more than one direction. CRC exposed an Accuracy–damage trade-off rather than a clear loss: multiscale mode narrowly led the primary scores, while FiberMargin corrected more corrupted labels and damaged more initially correct labels.

The external-prediction studies define the boundary of that evidence. FiberMargin improved both saved upstream predictions, but harmonic propagation and GraphST correction provided stronger operating points. The auxiliary binary ballot was also not the strongest rule on held-out building masks. These outcomes do not negate the multiclass enclosure result; they show that regular binary masks and errors inherited from an upstream predictor can favor simpler local operators. Recovery and protection must therefore be reported jointly, and method choice should follow the geometry and error regime rather than a universal ranking.

6.1 Scope of the mask-repair task

The same input contract covers regular image masks, 3D voxel labels, irregular point clouds, and categorical maps, but the present evidence is strongest for multiclass irregular-coordinate fields. It is most relevant when an upstream label field exists but its image, logits, training pipeline, or domain-specific model cannot be used. Potential tests include LiDAR labels (Ye et al. 2021), land-cover maps (International Society for Photogrammetry and Remote Sensing n.d.), and multi-annotator medical masks (Zhang et al. 2023). The supplement prespecifies suitable datasets and guards.

The taxonomy in Section 2 also clarifies how future comparisons should be organized. Single-mask, coordinate-only operators can be compared directly at equal information. Feature-guided refiners should form a second track that receives the image or point features; noisy-label methods should form a training-time track; and STAPLE-like methods should form a repeated-annotation track. A topology-aware track should report connectivity metrics and receive the structural information its methods require. SegRefiner and RNCA can learn visual or anatomical repair priors from data; FiberMargin cannot. Conversely, FiberMargin requires no fitted weights and applies unchanged to irregular 2D/3D coordinates. Collapsing these settings into one ranking would measure access to information as much as repair quality.

6.2 Limitations and next steps

The procedural masks, auxiliary HOT study, and public-annotation diagnostics use injected defects. CamVid and S3DIS instead supply genuine upstream predictions with independent references, but two simple upstream models cannot establish uniform performance across natural error mechanisms. The thin-network and coherent-patch failures show that the operator is not topology preserving. A contiguous regional overwrite can remain observationally ambiguous from geometry and one categorical field alone. The conditional recovery theorems require chart-interior witnesses and independent bounded label noise; they do not establish universal consistency at boundaries or under structured regional corruption. The finite atlas is not exactly rotation invariant.

The biological tests also remain controlled recovery studies. DLPFC contains several donors, but CRC and MERFISH each contribute one biological unit; natural annotation uncertainty was not independently adjudicated. Strong next tests require independent tissues, additional paired imperfect/reference masks from upstream models, and 3D point clouds. A feature-enabled veto should be evaluated separately from the present mask-only operator. For current evidence, FiberMargin is a fast, auditable multiclass spatial-label repair method whose strongest support is irregular controlled recovery; external-prediction generalization remains an open problem.

7 Software and Reproducibility

The R package exposes `clean_categorical_mask(mask, samples = NULL, workers = NULL)` for pixel/voxel arrays and `refine_spatial_labels(xy, labels, samples = NULL, workers = NULL)` for irregular coordinates. `corrupt_categorical_mask()` and `evaluate_mask_cleaning()` implement the public benchmark contract and exact repair-damage audit. The mathematical operator, including path construction, directional recurrences, margin aggregation, and decisions, is implemented in C++17. R validates and encodes inputs, allocates one total CPU budget, and restores output types and row order. One native call performs per-specimen axis reduction, coordinate normalization, local class encoding, and refinement. Specimens reuse the same budget and native stages do not nest thread pools. Independent charts and row ranges use deterministic standard C++ threads on macOS, Linux, and Windows. The returned factor retains input levels and identifiers and carries the seven pointwise diagnostics together with per-specimen dimensionality, sample size, edit count, edit fraction, before/after class sets,

and explicit removed-class summaries. The implementation does not constrain the number of output classes. The repository contains the versioned source, simulators, evaluators, manifests, scenario metrics, figures, and all scripts used here. Compact DLPFC and CRC coordinate/annotation derivatives are redistributed under their source terms. The CRC derivative is CC BY 4.0, records attribution and modifications, and excludes expression counts and tissue imagery. MERFISH source data are not redistributed; scripts document the official source and license. The HOT and Oxford-IIIT Pet controlled studies use mask labels only. The CamVid preparation script uses source images solely to create fixed upstream categorical predictions; the R repair evaluator reads only those saved masks, coordinates, and independent references, never image values or model scores. Its manifest records source terms and the exact upstream recipe.

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